



Clinical Considerations for Common Metabolic Diseases of Bovines

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“Learning is the continuous process”



Common Metabolic Diseases in Bovines

- Ruminal Acidosis
- Hypocalcemia
- Hypophosphatemia
- Hypokalemia
- Downer Cow
- Bovine Ketosis
- Hypomagnesaemia

Clinical Consideration in Ruminal Acidosis

- Pathophysiology
- Diagnosis
- Early the Treatment –Faster the recovery---**HR is prognostic Indicator**
- **Primary Treatment**---Sodium Bicarbonate IV in NS–dose Calculation

BASE DEFICIT SCORE

Dose Calculation of Sodium Bicarbonate

$$0.3 * \text{Base deficit} * \text{body weight} = \text{meq of NaHCo}_3$$

BASE DEFICIT

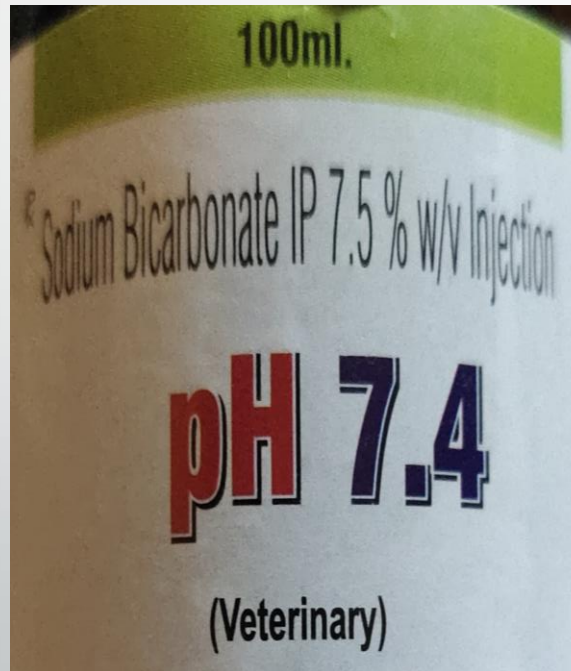
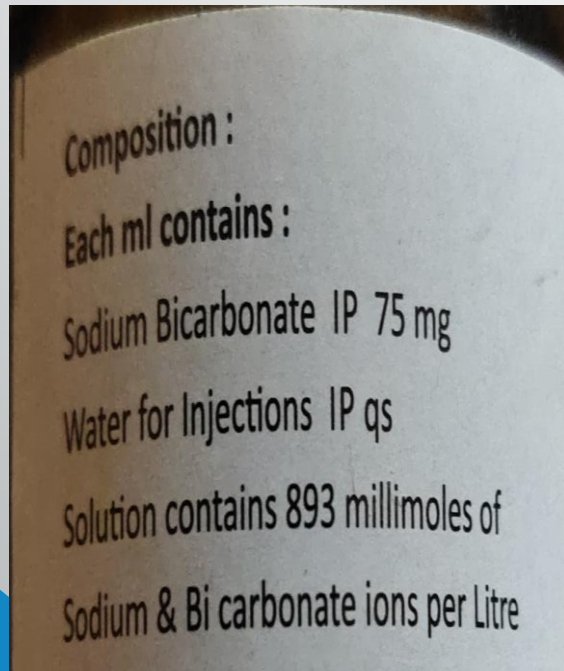
- **MILD(5)**-Animal is walking, Fluid Splashing Sound on Rumen Palpation, SemiSolid Dung, Inappetance
- **MODERATE(10)**-Incoordination, Staggering Gait, Diarrhea, Dull and Depressed, Respiratory Distress, Complete Anorexia
- **SEVERE(15)**-Sternal/Lateral Recumbency, Increased Heart Rate

Dose of Sodium Bicarbonate –Mild/Modeate/Severe

- Method 1

$0.3 \times \text{Base deficit} \times \text{body weight} = \text{meq of NaHCO}_3$

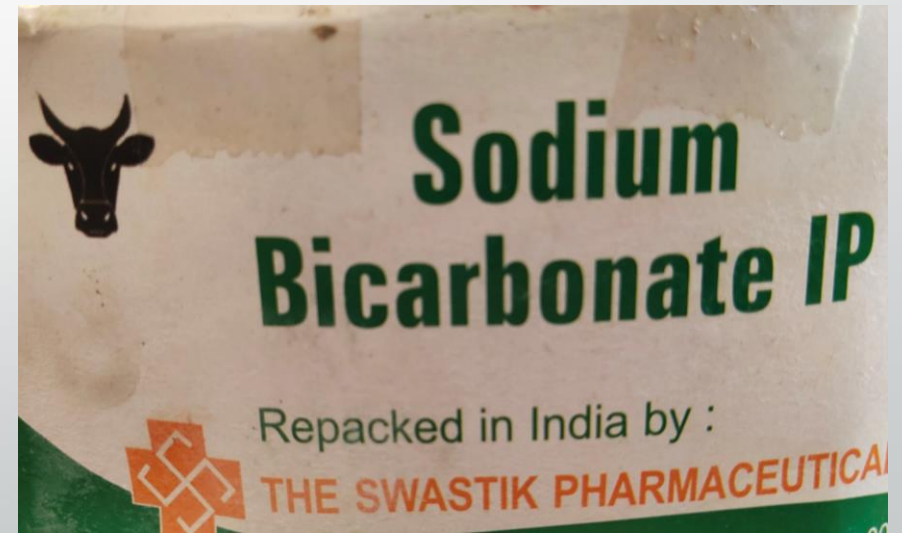
Each ml has 0.9meq of NaHCO₃ or each ml has 75mg



- Method 2

- Thumb Rule

- For 500kg cow, Adm 125g of Sodium Bicarbonate in 10litres of isotonic solution Iv for every 2% dehydration-10-20mins



Clinical Consideration in Ruminal Acidosis

- **Adm Of oral Antacids**-Sodium Bicarbonate/Aluminium Hydroxide/Magnesium Hydroxide @ 1g per kg bt wt-5litres of water
- If IV Sodium Bicarbonates are used,,it is undesirable to adm oral Antacids

Clinical Consideration in Ruminal Acidosis

- Ab targeting Gram positive S.bovis esp..**Procaine Penicillin G**, OTC or ST
- NSAIDs -Flunixin for Endotoxemia, Antihistamines and Thiamine @ 10mg per Kg q24hrs for 3days
- Fluid Therapy for Rehydration-RL and NS---Strictly No DEXTROSE
- Acidosis induced Hypocalcemia-Atonic Rumen-CBG SC/IV

Hypocalcemia(Milk Fever)

- Etiology—Transition Period
- Differential Diagnosis-1)Ruminal Acidosis
2)Endotoxemia
3)Calving Paralysis
- Ruminal Acidosis--Rumen palpation -Fluid Filled , Dung –Semi Solid
Respiration- Shallow
- Milk Fever- Rumen will be Doughy, Dung-Solid and Not voided,Pupil dilated,Relaxation of Anal Sphincter.

Clinical Consideration in Milk Fever

- Dose of Calcium Borogluconate in MF is 2.2gms per 100kg (350kg-7.7g)
- 450ml of IV CBG available provides 8.37g of Calcium
- Warm the bottle and Give IV-Two Reason-1)Normal BT 2)Faster Assumulation and Absorption
- Half of the dose IV and Balance SC
- IV ca will be available in Blood Stream only for 6-8Hrs

Clinical Consideration in Milk Fever

- Always Prefer Plain CBG in Milk Fever Than others.
- SC injection of Ca per site -50-125ml (Average -75ml per site)-don't press/Massage
- Minimum Calcium IV dose adm rate is 15mins-to avoid Recurrences.
- Fast Adm leads to Cardiogenic Shock as Ca is CardioToxic
- Don't Go for Oral Calcium preparation when the animal is in Sternal or Lateral Recumbency-Aspiration Pneumonia

Clinical Consideration in Milk Fever

- Do not Adm more than 2 Bottles of CBG in a case of Recumbent Cow Not responding to CALCIUM

When Not to give CALCIUM

- Ruminal Acidosis
- Peritonitis
- Septicemia- Acute Mastitis
- Toxemia
- When Raise in Temperature

Clinical Consideration in Hypophosphatemia

- Diagnosis- Relapsing /Recurrent Animal
- TWO TYPES OF AVAILABLE PHOSPHOROUS in the Market
- In Bovines, The Blood phosphorous level is direct indicator of Dietary intake and **Bioavailability of Phosphates**(Inorganic) and it is also a factor for Bovine Recumbency (Recurrence/Persistent)
- INJECTABLE/ORAL-ONLY PHOSPHATES(INORGANIC)
- Dose of Inorganic Phosphorous -60g of SAP in 200ml NS/Distilled water IV
-60g of SAP in 200ml NS/Distilled water SC 12hrs interval
oral-120g of SAP bid

Clinical Consideration in Hypophosphatemia

- It is **undesirable** to adm parentally Sodium Acid Phosphate(SAP) along with Calcium and Magnesium salts **IV-Rapid infusion-precipitation** of Ca and Mg salts
- IV adm of **PHOSPHATES** can be done after 2hrs where the animals not able to take oral Medications
- **Rapid Infusion of Phosphates** –low bioavailability-after 2hrs-it will be excreted via Kidneys-so little parenteral usage only

Clinical Consideration in Hypophosphatemia

- Phosphorus depletion may **ALSO occur** after oral or **parenteral carbohydrate adm** and after **parenteral insulin administration** as a result of increased cellular phosphorus uptake in combination with glucose.

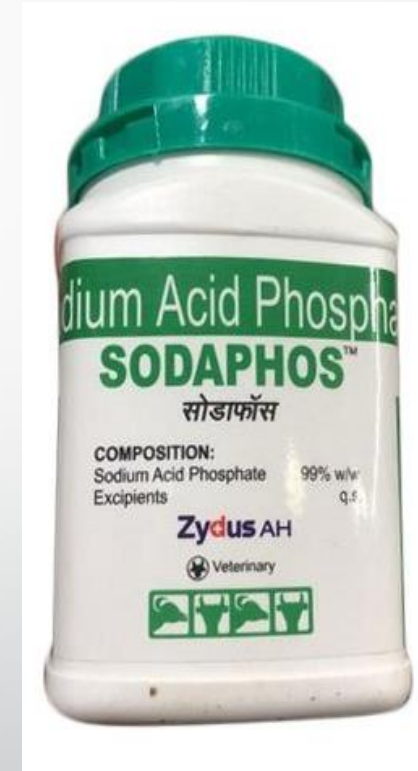
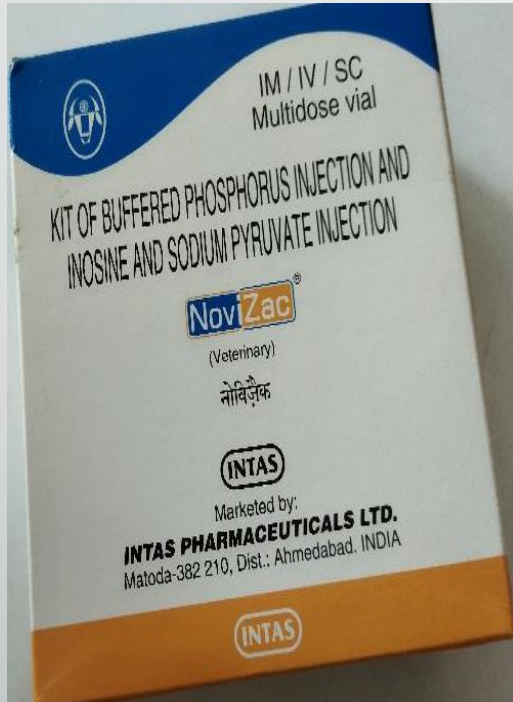
PHOSPHITES(ORGANIC)

- Parenteral adm of **organic phosphorus** (inj toldimphos, butaphosphan, phosphite, or hypophosphite) as **phosphite** are unsuitable for increasing plasma phosphorus.
- **PHOSPHITES- Metabolic Modifiers**-can be used in other conditions like SI and Ketosis

Inorganic phosphorous Vs Organic phosphorus

- **INORGANIC PHOSPHOROUS-as phosphates**
- **INJ NOVIZAC 50ml** (INTAS)-as Phosphate- Single dose for 450kg btwt IV
- **INJ URIMIN** - (SAPas ip8%w/v) (VIRBAC)- 50ml 3 doses 12hrs interval IV
- **INJ ALPHOS-40**(SAPas ip 8%w/v)
- **Oral-Sodaphos (SAP)**-zydus-50g bid
- **Oral P-Soda** (SAP)-Hexter-50g bid
- **ORGANIC PHOSPHOROUS-phosphonic acids and phosphan form**
- **INJ TONOPHOSPHAN VET-MSD** (SS4DM2MPPA as organic p 20%)
- **INJ T-PHOS**(SS4DM2MPPA as organic p 20%)-zydus
- **INJ CATOSAL**(Butaphosphan +B12)-bayer
- **INJ INJECTIPHOS**(Butaphosphan +B12)-zydus
- **INJ SYNKOMET** (Butaphosphan +B12)- intas

Kits for Phosphorous supplementation



Clinical Consideration in Hypokalemia

- **Prolonged Recumbency**—Increase in permeability of Sarcoplasmic membrane-loss of K from intracellular to Extracellular –Excretion via Kidneys—Hypokalemia (**andrew et al, 1992**)
- **Muscle weakness** –Lowers resting membrane potential – decreased Excitability of NM tissues-Downer Cow syndrome (**Radostits et al,2000**)

Other Cdts causing Hypokalemia

- Over dosage of **Bicarbonate solutions** in Acute Ruminal Acidosis-Alkalosis-Renal Excretion

Clinical Consideration in Hypokalemia

- **Multiple dosing of Isofluperidone** in Ketosis-Mineralocorticoid activity-Renal Excretion
- Fluid Therapy in diarrhea
- Enteritis-**ABOMASAL STASIS**-Hypokalemia

Dosage calculations-parenteral Vs oral

- **Oral safe**-0.4g per kg-bid ie 120g bid-severe Hypokalemia
- **Injectable-unsafe**-only after Lab investigations-cardiac arrhythmias-VF and Death.

Available potassium kits



Lab Grade KCL 500gms-
MRP 282



200ml has only 20g-
MRP 70rps

ORCALYTE-DOWNER

A Vital Supplements for Downer Cow





COMPOSITION :
 Each Packet Contains :

- Sodium Chloride : 45.0 gm (770 mmol)
- Potassium chloride : 180.0 gm (2410 mmol)

 (Total Osmolarity 3180 mOsmol/L of water and 132.5 mOsmol/ 24 L of water)

DIRECTION FOR USE :
 Dissolve entire contents of the packet in 24.0 Litres of normal drinking water and give 2 times over a period of 24 hrs.

Unused diluted water should be discarded after 24 hours of reconstitution.

RECOMMENDED FOR :
 Downer cow syndrome.
 Hypokalemia.

DOSE :
 As directed by the veterinarian.

STORE IN A COOL & DRY PLACE. PROTECT FROM DIRECT SUNLIGHT
 FEED SUPPLEMENTS, NOT FOR MEDICINAL USE
 KEEP OUT OF REACH OF CHILDREN
 NOT FOR HUMAN USE, FOR VETERINARY USE ONLY.

Batch No : OOD - 01
 Mfg. Date : APR 2019
 M.R.P : 120/-
 Exp. Date : 36 months from mfg. date.

Orcalyte Downer-MRP 120

Take Home Points

- Ruminal Acidosis
- Hypocalcemia
- Hypophosphatemia
- Hypokalemia



End Discussion.....

Thank you

Spl Thanks to Respected Dr S.Sivaraman Sir